Tree Borrows

Neven Villani,[1] Johannes Hostert,[2] Derek Dreyer,[3] Ralf Jung[2]

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[1]Univ. Grenoble Alpes, Verimag
[2]ETH Zurich
[3]MPI-SWS

Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
  *x = 13;
  *y = 20;
  *x
}
```

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mutable thus disjoint

*x has known value

*x is unchanged

always returns 13

Rust's type system enables powerful optimizations

```rust
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
  *x = 13;
  *y = 20;
  13 // formerly *x: one fewer load from memory
}
```

Type-level guarantees for references

~~aliasing~~

~~&~~

~~mutability~~

`&mut` → mutation, no aliasing

`&` → aliasing, no mutation

Escape hatch: `unsafe`

Can use **unchecked operations** to do **low-level manipulations**

```
unsafe {
  // Code within this block can effectively
  // bypass some parts of the typechecker.
  ...
}
```

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- absence of data races
- ...

What if `unsafe` code is misused ?

```rust
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
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}

fn main() {
  let mut root = 42;
  let ptr = &raw mut root;
  let x = unsafe { &mut *ptr };
  let y = unsafe { &mut *ptr };
  println!("{}", write_both(x, y)); // prints 20
}
```

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UB: Not the compiler's responsibility

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
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- absence of data races
- compliance with aliasing rules NEW!

violations trigger UB

Tree Borrows (TB): defines those aliasing rules

Compiler **assumes absence of UB**, exploits this for optimizations

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weak optimizations

hard to write correct code

less UB

more UB

UB: Not the compiler's responsibility

Within `unsafe` it is **the programmer's responsibility** to check

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- that memory is initialized

Sounds familiar?

Stacked Borrows has the same purpose,

Tree Borrows is its successor.

weak optimizations

hard to write correct code

less UB

more UB

Stacked Borrows (SB)

In safe Rust, the Borrow Checker makes borrows well-bracketed.

Stacked Borrows extends the well-bracketedness to unsafe.

```
let mut root = 42;
let ptr = &raw mut root;
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root

- new stack at `root`

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ptr
root

- ✓ `root` is at the top
- push `ptr`

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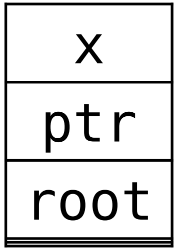

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x
ptr
root

- pop until `ptr` is at the top

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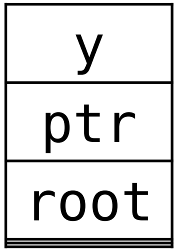

- pop until `ptr` is at the top
- push y

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y
ptr
root

- search for `x`

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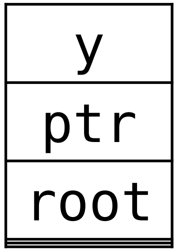
UB!



Can't use x if it is not in the stack

Desired outcome: UB

SB was **implemented** in Miri (official interpreter and UB detector)

→ included in many projects' CI

→ many bugs detected (e.g. in stdlib)

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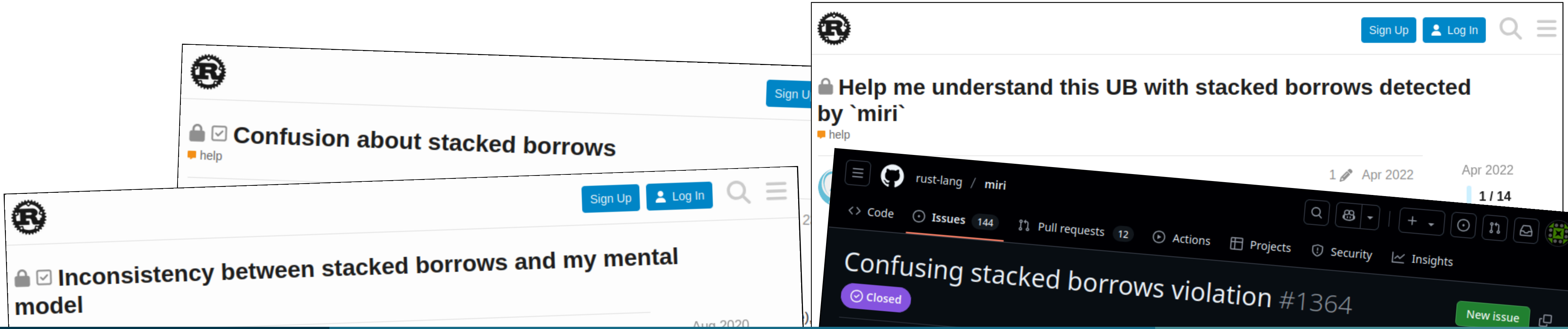

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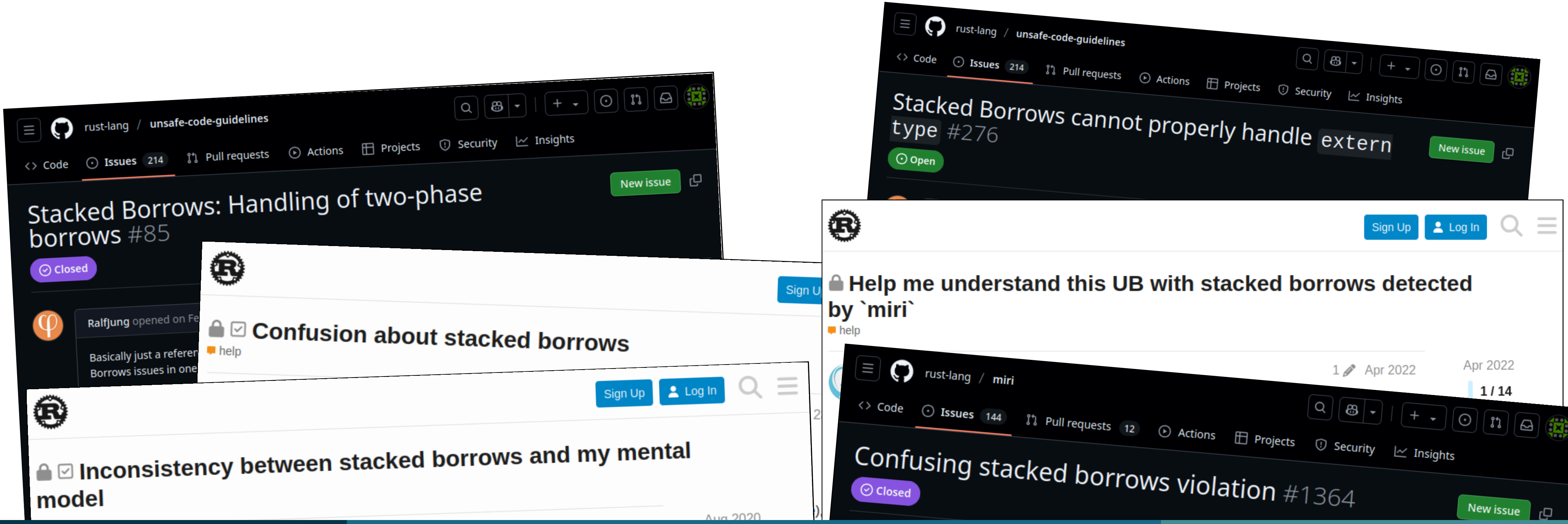

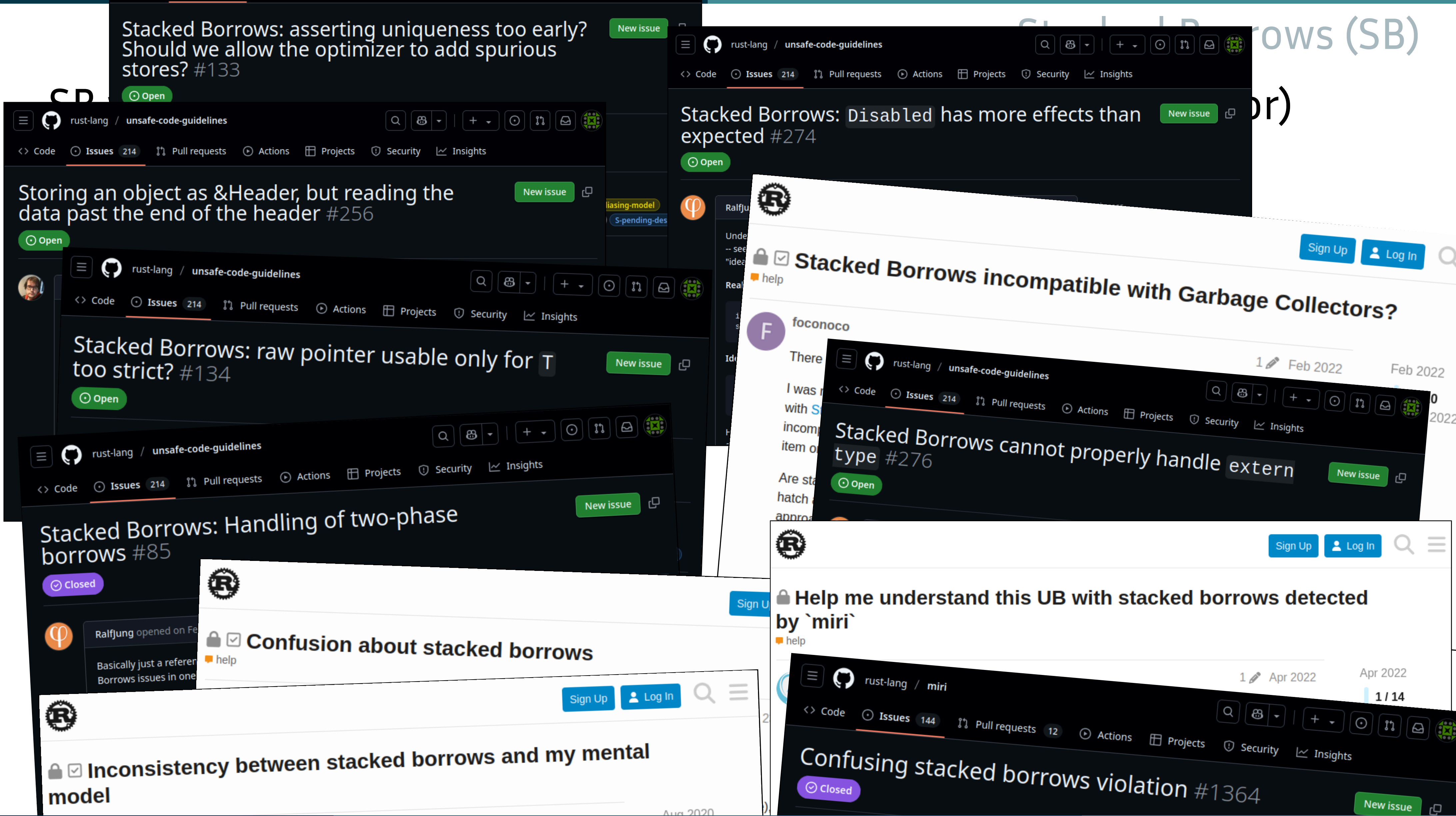
Stacked Borrows: asserting uniqueness too early? Should we allow the optimizer to add spurious stores? #133
Open
rust-lang / unsafe-code-guidelines
Stacked Borrows: Disabled has more effects than expected #274
Open
Storing an object as &Header, but reading the data past the end of the header #256
Open
Stacked Borrows: raw pointer usable only for T too strict? #134
Open
Stacked Borrows incompatible with Garbage Collectors?
help
foconoco
Stacked Borrows cannot properly handle extern type #276
Open
Stacked Borrows: Handling of two-phase borrows #85
Closed
Confusion about stacked borrows
help
Help me understand this UB with stacked borrows detected by `miri`
help
Inconsistency between stacked borrows and my mental model
rust-lang / miri
Confusing stacked borrows violation #1364
Closed

Anecdotal evidence supported by data:

- analysis of 30 000 libraries
- 6000+ tests have aliasing UB under Stacked Borrows (leading cause of UB)

Tree Borrows allows much more code

Tree Borrows uses a **tree** instead of a stack to track borrows

Out of 30 000 most downloaded libraries,
54% **fewer tests** with aliasing UB when using Tree Borrows

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fixes known technical limitations of SB,
incl. 2-phase borrows, extern types, **pointer offsets**

From Stacks to Trees

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let mut root = vec!['a','b','c'];
let x0 = &raw mut root[0];
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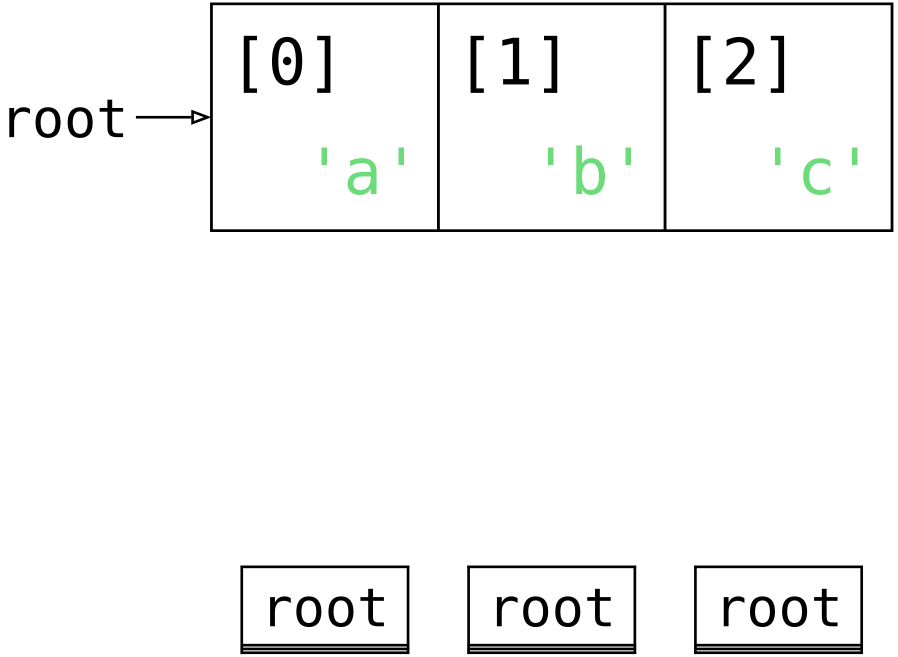

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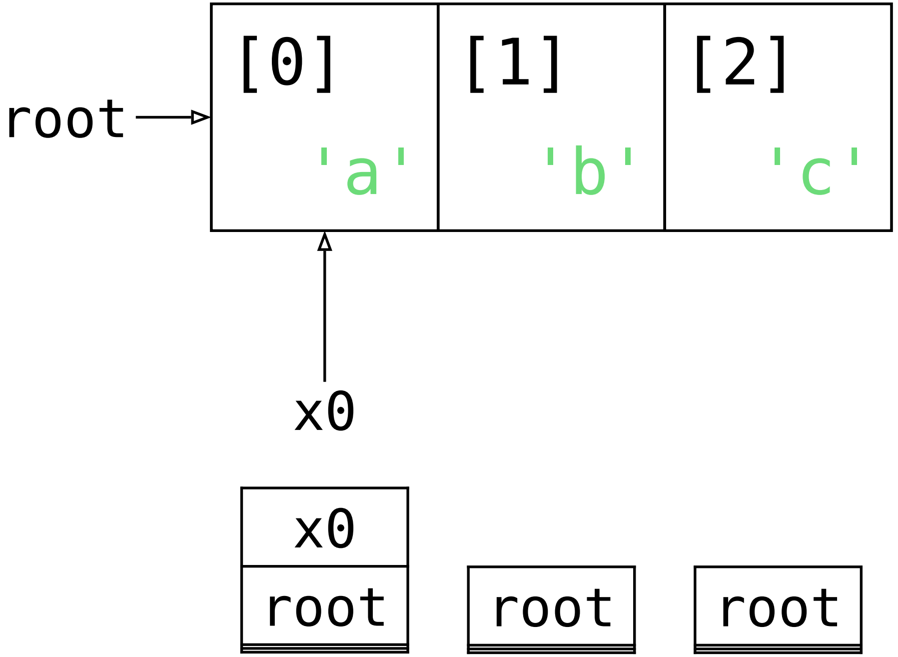

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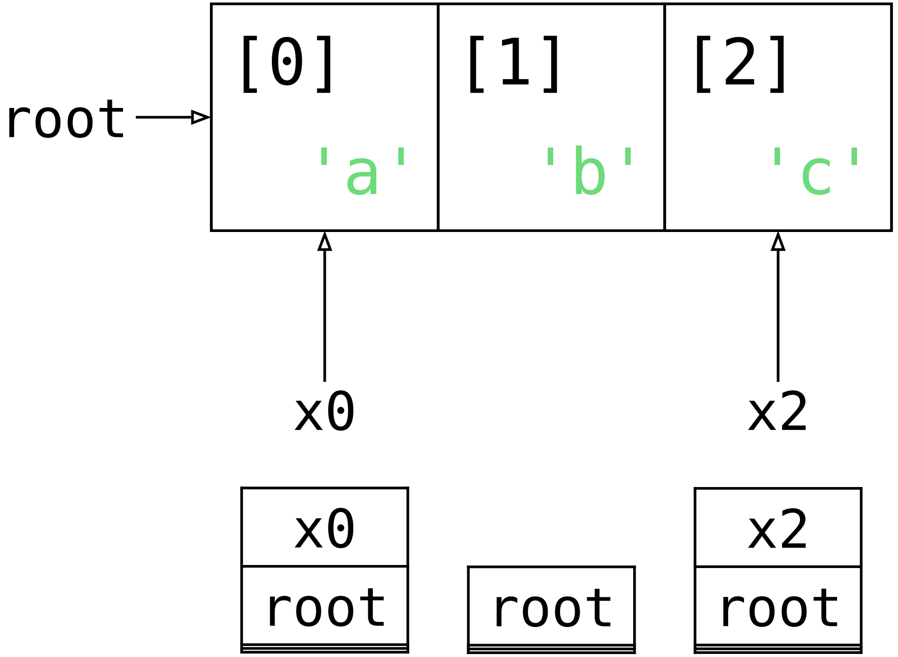

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// Scenario 1
unsafe { *x0.add(1) = 'z'; }
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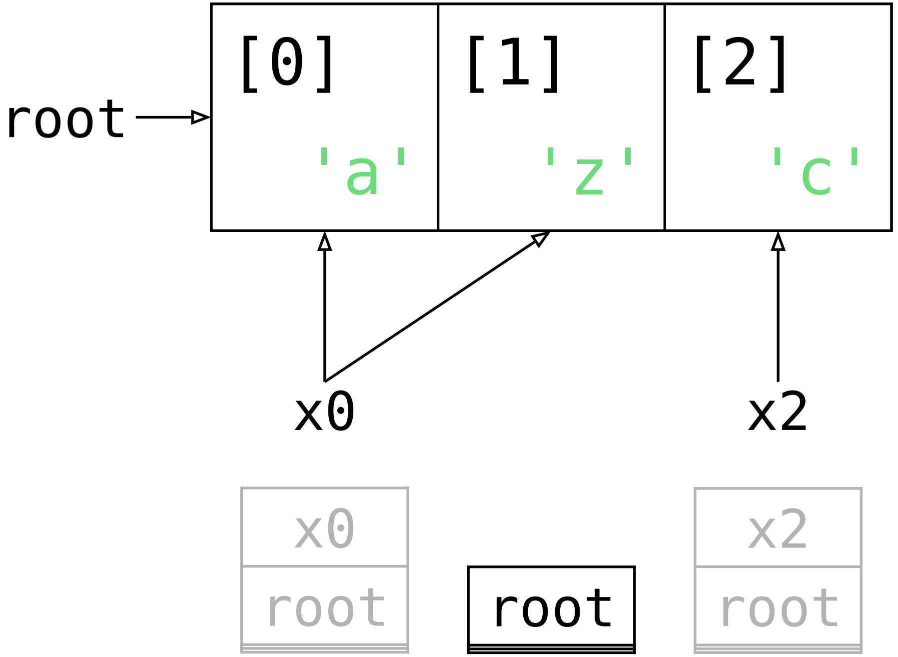

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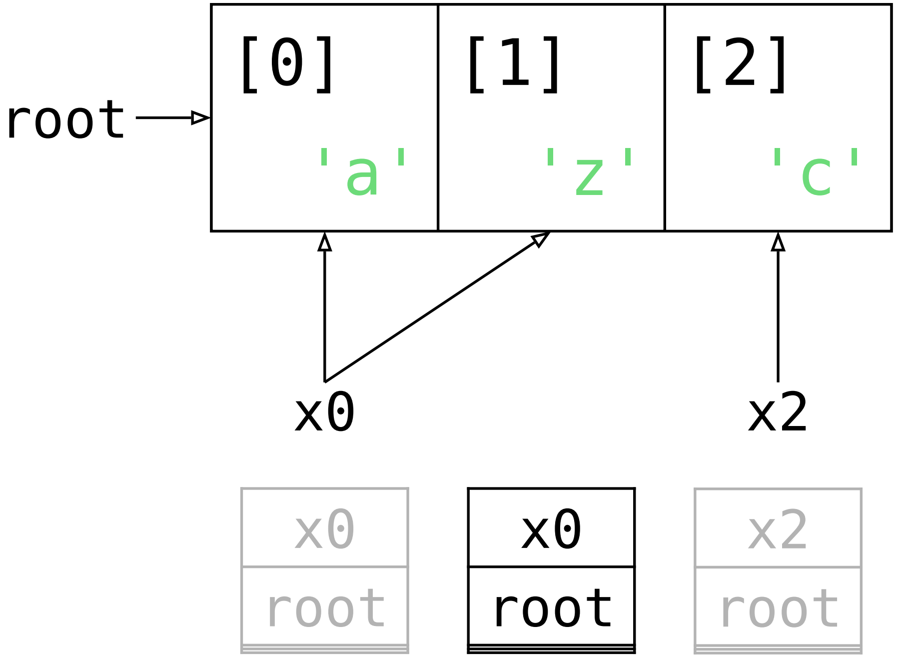

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// Scenario 2
unsafe { *x2.sub(1) = 'z'; }
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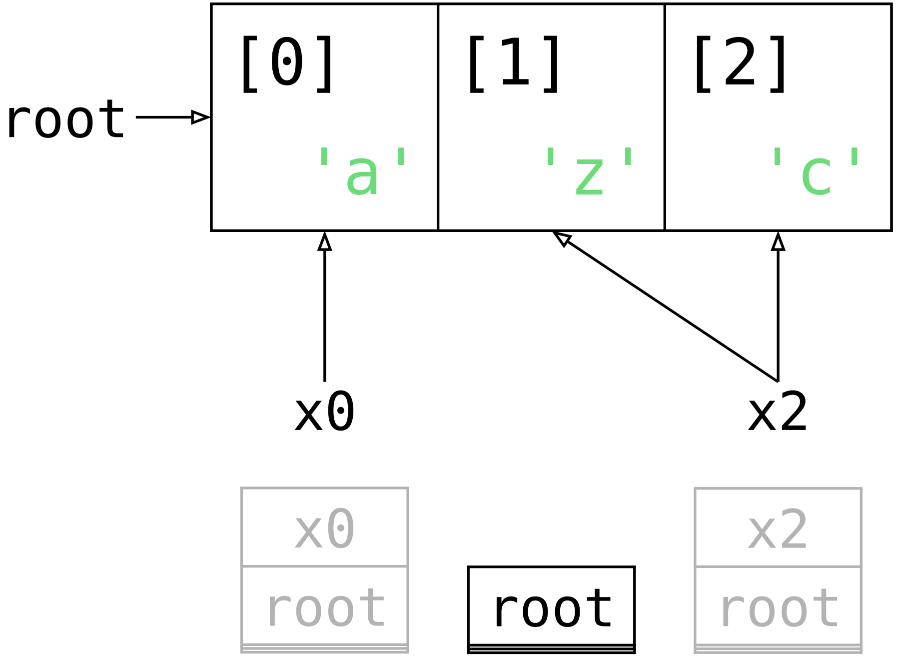

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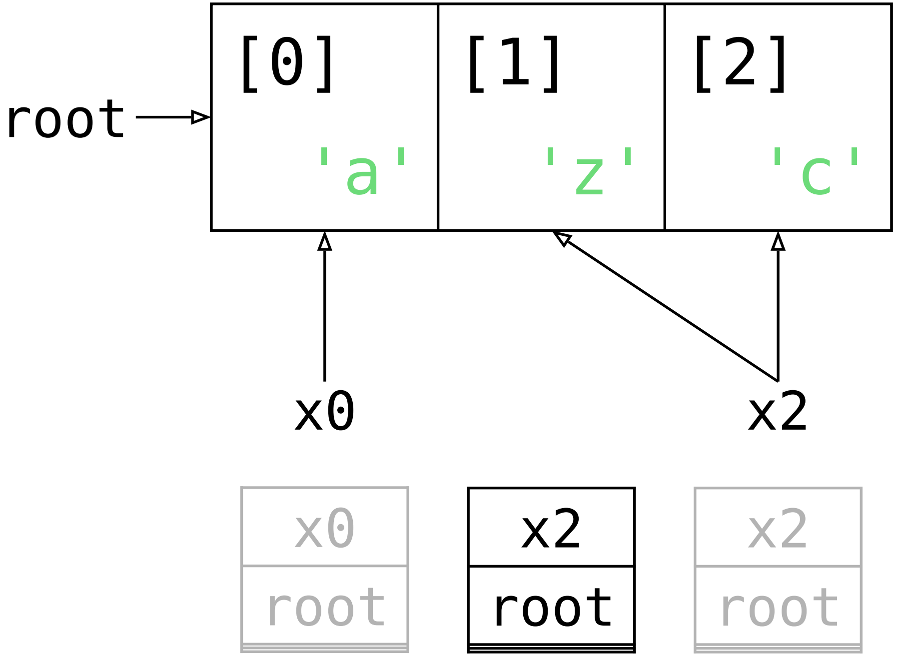

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// Scenario 1 or 2
unsafe { *??? = 'z'; }
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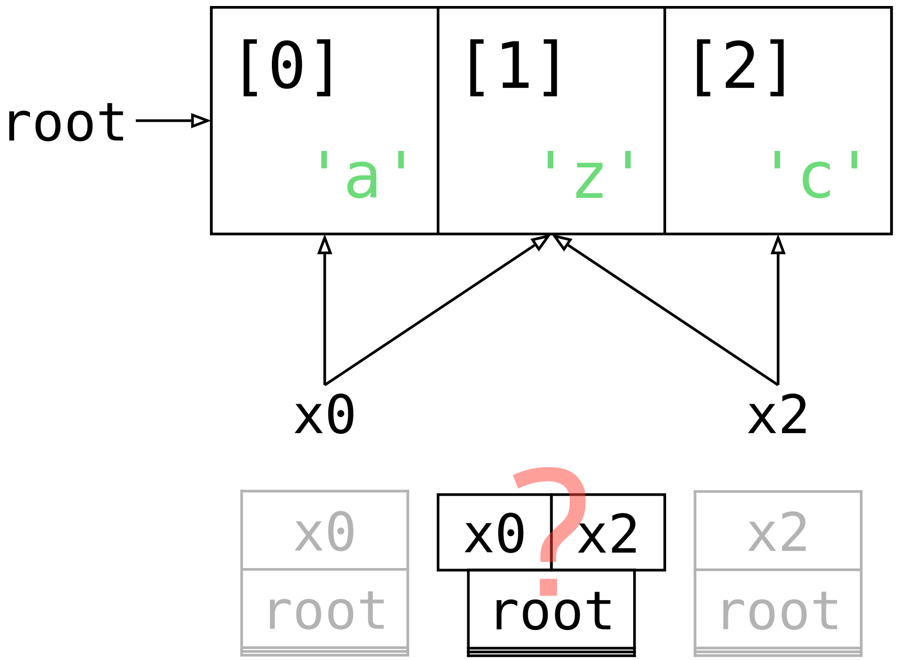

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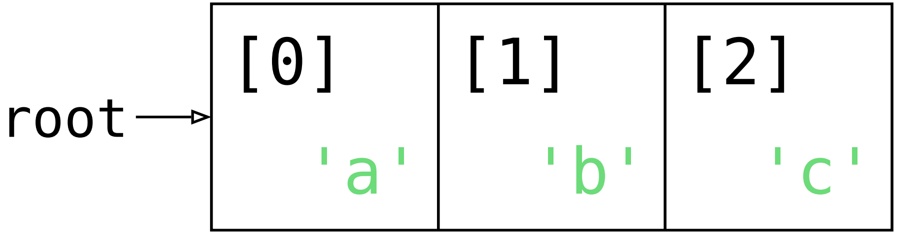

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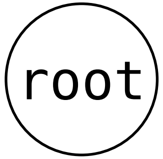

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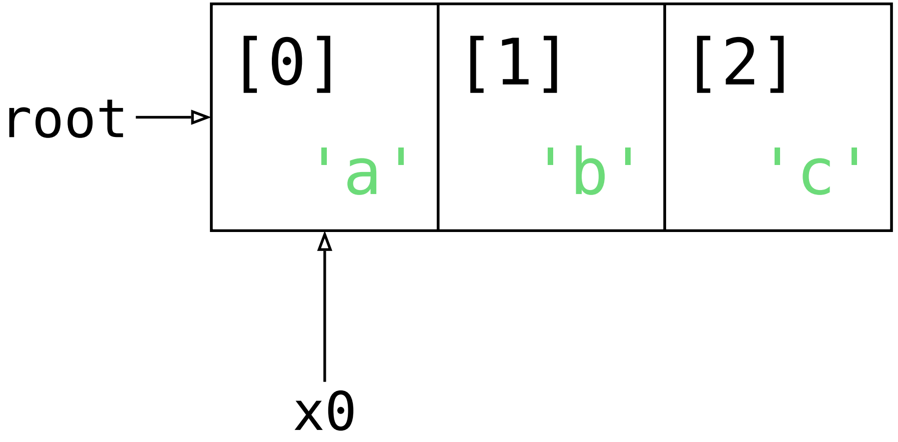

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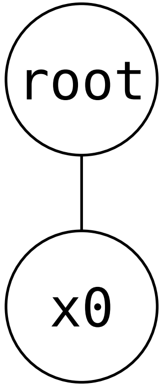

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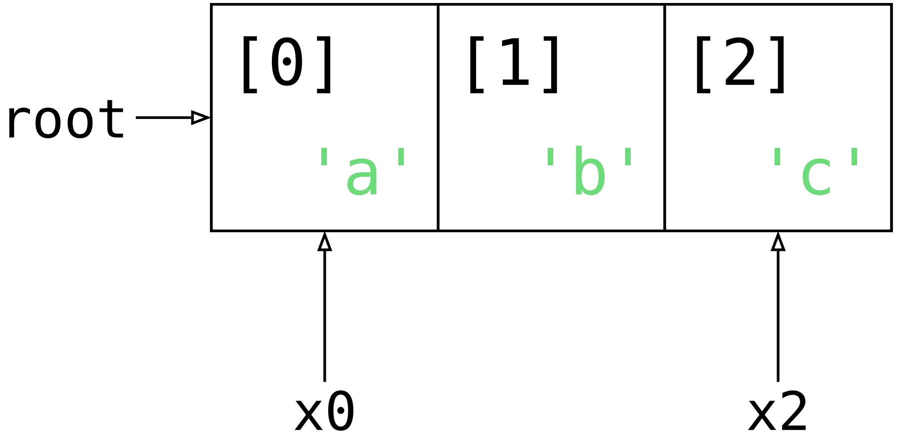

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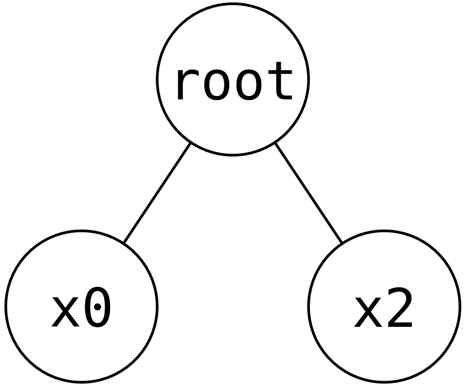

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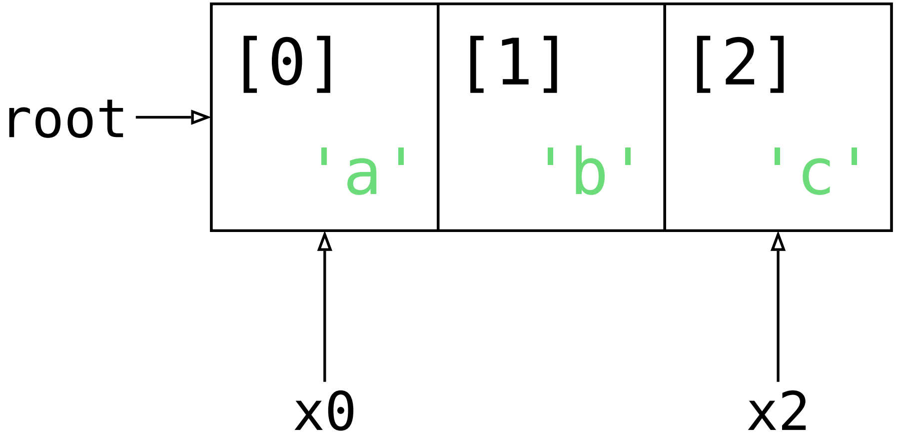

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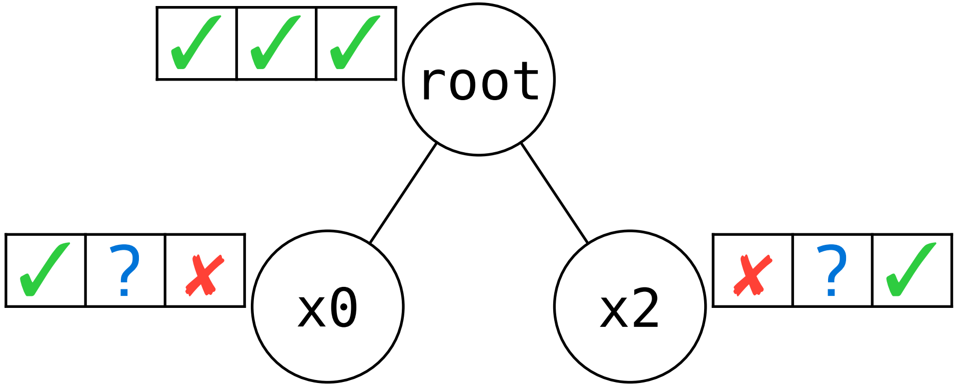

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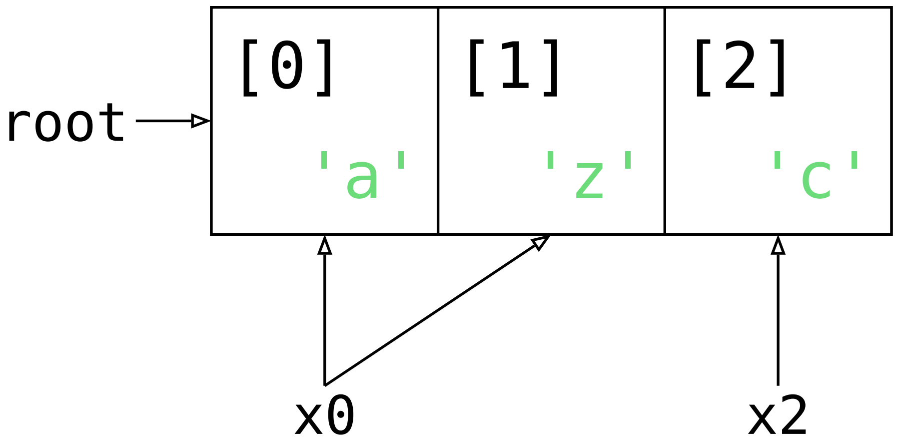

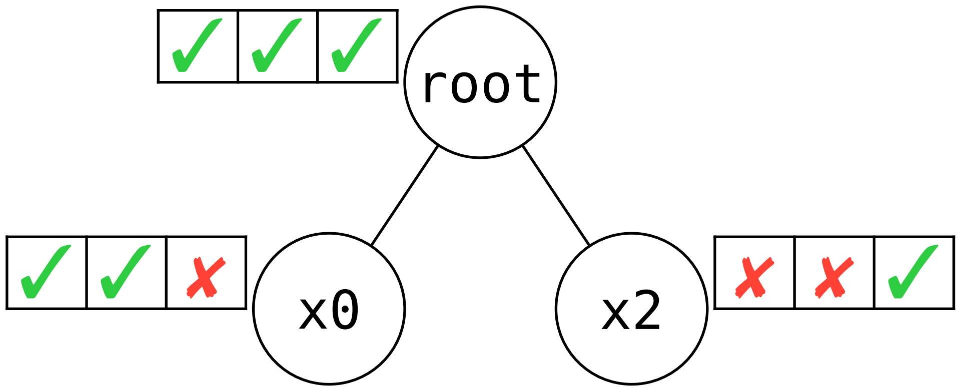

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// Scenario 2
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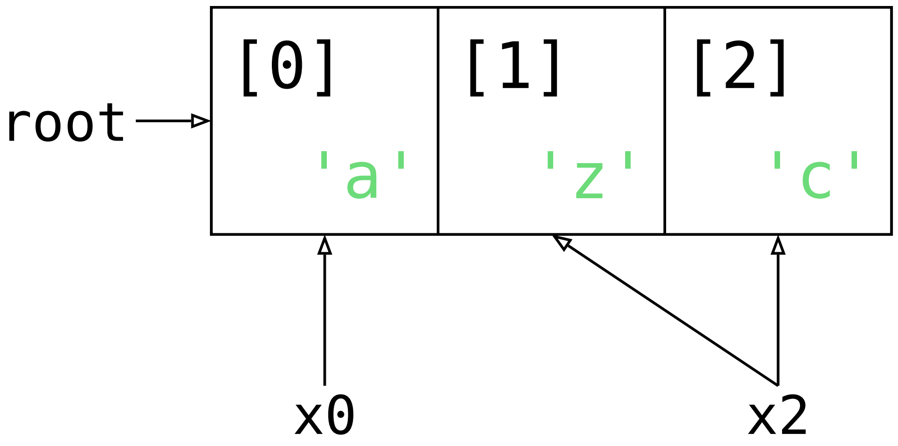

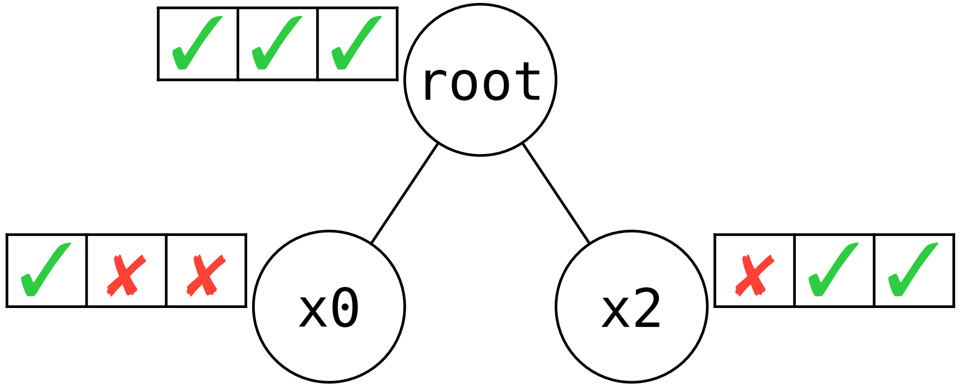

A second look at the motivating example


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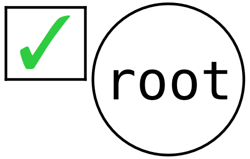

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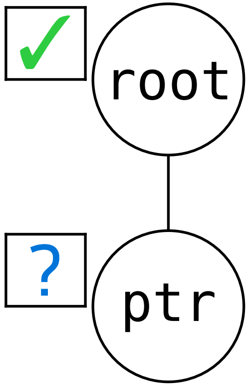

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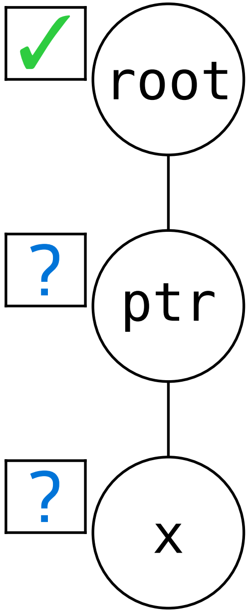

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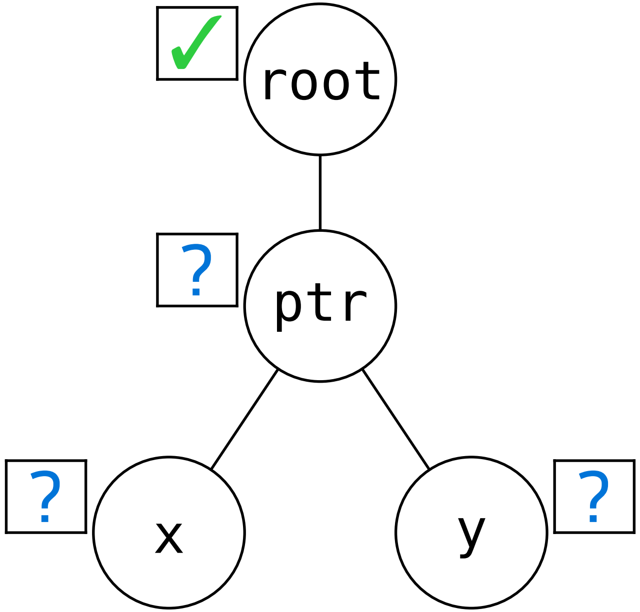

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```
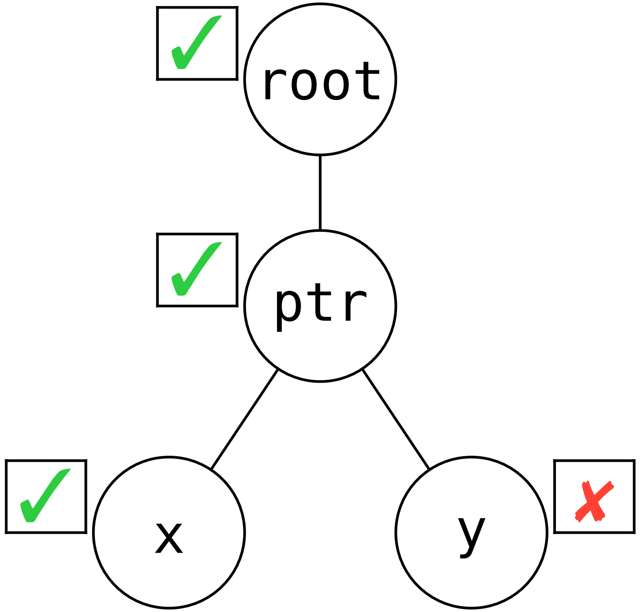
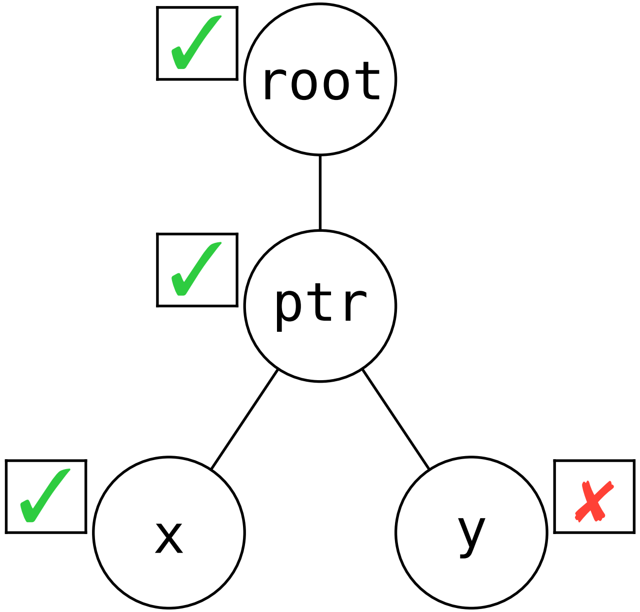



Desired outcome: UB

A second look at the motivating example

```
let mut root = 42;
let ptr = &raw mut root;
let x = unsafe { &mut *ptr };
let y = unsafe { &mut *ptr };
// inline write_both(x, y):
*x = 13;
*y = 20;
```

UB!



Desired outcome: UB

Evaluation

TB should enable desired optimizations

i.e. have enough UB to rule out problematic patterns

- formalized in Rocq (+Simuliris)
- a selection of optimizations proven
  - ✓ delete read through `&mut` or `&`
  - ✓ insert read through `&` in function
  - ✓ move read down for `&mut` or `&` in function

  …

It should be possible to write `unsafe` code free of UB

i.e. UB should be predictable and not too common

54% fewer tests have aliasing UB according to Tree Borrows

Only 31 ($< 0.01\%$) tests are regressions, all easily fixable.

(Out of 30 000 libraries, 400 000+ working tests)

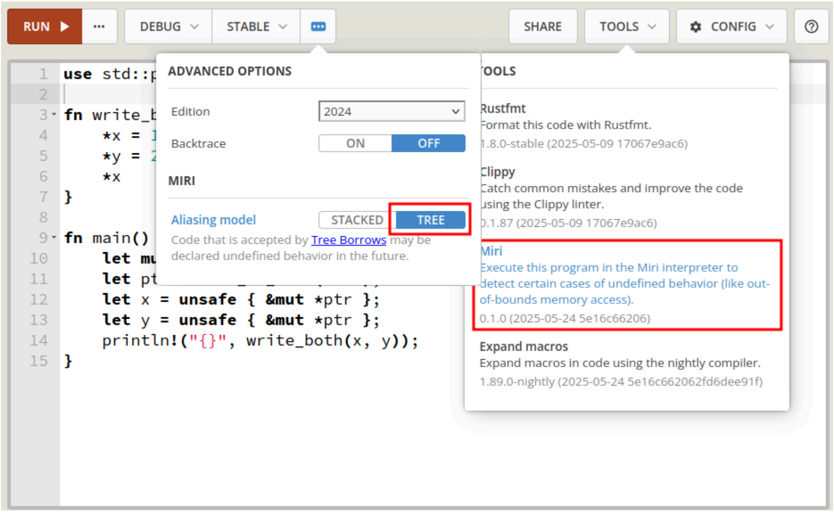
"Tree Borrows accepts more real-world programs that call foreign functions than Stacked Borrows due to differences in handling pointer arithmetic."

A Study of Undefined Behavior Across Foreign Function Boundaries in Rust Libraries, by I. McCormack, J. Sunshine, J. Aldrich @ ICSE'25

Conclusion

Try it out: supported by Miri

Also on the Rust Playground (`play.rust-lang.org`)



Postdoc positions available

@ ETH Zurich `ralf.jung@inf.ethz.ch`

@ MPI-SWS `dreyer@mpi-sws.org`

Learn more:

`plf.inf.ethz.ch/research/pldi25-tree-borrows.html`

Includes e.g. handling of raw pointers and interior mutability.